# A Rank-1 Suppression Direction Mediates Prompted Sandbagging in Gemma 2 2B IT

*Causal mechanism, neighborhood geometry, and the limits of probe and SAE-feature proxies at* `attn_out_L12`

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**Abstract**

Capability evaluations gate deployment of contemporary language models, but instruction-tuned models can detect that they are being evaluated and selectively underperform on the evaluation itself. We present a circuit-level causal characterization of this phenomenon in Gemma 2 2B IT and use the identified mechanism as a fixed causal anchor against which neighborhood geometry, cross-scale transfer, and standard SAE-feature selection are measured at the same site.

The headline mechanism is a rank-1 direction at the attention output of layer 12 ($d_{\text{mech}}$). Computed as the sandbag–honest difference-in-means at the prompt-final token and applied as a single-fire projection-out on the first forward pass, $d_{\text{mech}}$ restores sandbag-condition accuracy by +5.0 / +12.5 / +10.0 percentage points on val / test / fresh held-out prompts ($n = 40$ each), with 0 pp damage to honest fresh accuracy and 0 pp from a random-direction baseline. SVD confirms precisely rank-1 structure ($\sigma_2/\sigma_1 = 0.275$). The direction is near-orthogonal to refusal (cos = 0.357 at the same site; the refusal-orthogonal component recovers the full effect), is dominated by an eval-detection signal rather than a separable explicit-underperformance signal, and transfers across-condition to paraphrased and user-turn-injection trigger families at +10 pp restoration each.

We embed $d_{\text{mech}}$ in a three-stage vertical mechanism. Activation patching localizes the upstream decision to layer 11's residual stream (peak $\Delta_{\text{proj}}$ 32.4%, +20 pp behavioral, trigger-invariant; single-session result, disjoint-prompt replication registered as future work); direct logit attribution identifies layers 8–10 attention as the generators and layer 11 as the carrier. Downstream layers L13–L25 partially compensate through orthogonal subspaces, restoring approximately 51% of the suppression-direction-aligned component while growing the residual-stream perturbation $\sim 3\times$ in directions orthogonal to $d_{\text{mech}}$.

Three neighborhood characterizations sharpen the picture at the same site. (i) Composition with refusal, probe, and honesty directions is non-additive (5 of 10 pairs deviate from summation; 3 of 3 OVERRIDING pairs include the eval-detection axis), and dosage on ($d_{\text{refusal}}$, $d_{\text{mech}}$) saturates non-monotonically at $\alpha \approx \|r\|$. (ii) The analogous direction in Gemma 2 9B IT, geometrically aligned with $d_{\text{mech}}$ at cosine 0.921 (eval-detection) and 0.996 (sandbag) after PCA-200 Procrustes alignment, produces no causal effect at the corresponding layer; geometric universality across model scales does not imply causal universality. (iii) Mean-shift SAE feature selection at $K = 20$, applied at the same site, fails to recover the mechanism under both cross-stage (base $\rightarrow$ IT) and within-condition contrasts: the cross-stage $K = 20$ ablation produces $\Delta_{\text{count}} = 1/200$ against

a pre-registered threshold of $\geq 20$ despite 2594 features clearing $2\sigma$ at the L12 contrast; the within-condition $K = 20$ ablation produces an opposite-signed $-15$ pp effect on the same $n = 40$ test split where $d_{\text{mech}}$ recovers $+10$ pp.

The unifying observation is that the sandbagging mechanism at L12 is geometrically simple and causally narrow, and that simpler proxies—neighbor steering directions, scale-aligned directions, and top-$K$ statistical features—locate representations correlated with but causally distinct from it. The L12 direction is a candidate post-hoc activation-level defense bounded by three negatives (asymmetric polarity, sparse selectivity, variant-dependent behavior under conflict prompts) and a fixed anchor against which probe-direction and SAE-feature methods can be benchmarked.

## 1 Introduction

A capability evaluation that the evaluated model has already detected as a capability evaluation is no longer a capability evaluation. Instruction-tuned language models can be prompted, fine-tuned, or password-locked to selectively underperform on evaluations while maintaining performance on harmless prompts [27]; the resulting behavior is detectable by linear probes on residual-stream activations at high AUROC [9]; and the evaluation-awareness signal that drives much of the effect scales predictably across major open-weight families [4]. The detection-oriented line of work has been complemented by adversarial stress-testing of probe-based schemes [23] and by demonstrations that probes combined with SAE-feature steering can partially recover sandbagged capability on Llama-3.3-70B [17].

Detection identifies *when* a model is sandbagging; it does not by itself identify the computational substrate that produces the behavior, nor does it tell a deployer whether targeted intervention at that substrate would restore capability without compromising other model properties such as refusal of harmful instructions. This paper provides a causal counterpart for Gemma 2 2B IT [8] and then uses the identified mechanism as an anchor to ask what other interpretability instruments—steering-vector composition, cross-scale direction alignment, SAE-feature selection by mean activation difference—actually capture at the same site.

**Contributions.**

1. **A validated rank-1 intervention object at `attn_out_L12`.** The sandbag–honest difference-in-means direction at the prompt-final token, computed on train $\cup$ val and L2-normalized, restores sandbag-condition accuracy by $+5.0$ / $+12.5$ / $+10.0$ pp across val / test / fresh held-out prompts ($n = 40$ each; single test-touch consumed) with 0 pp damage to honest fresh capability. SVD confirms precisely rank-1 structure ($\sigma_2/\sigma_1 = 0.275$). The intervention is a single 2304-dimensional unit vector applied at one component on one forward pass.

2. **A three-stage vertical mechanism characterization.** Activation patching at `resid_pre[L11]` reduces the L12 sandbag-direction projection by 32.4% with $+20$ pp behavioral capability restoration on the discovery set (single-session STRONG-POSITIVE; replication on disjoint prompts registered as future work), replicating across two alternative sandbagging triggers. Direct logit attribution identifies L8–L10 attention as the top-3 contributors to L12's sandbag-direction signal; L11's own attention output contributes only $+0.044$. Downstream layers L13–L25 partially compensate through orthogonal subspaces.

3. **Mechanism properties at L12:** precisely rank-1; refusal-orthogonal (Jaccard 0.042 with refusal at L15 SAE features, cosine 0.357 with refusal at L12 `attn_out` across two independent levels);

eval-detection-dominated rather than dissociable into a separable explicit-sandbag direction; trigger-invariant across-condition for two alternative sandbagging trigger families.

4. **Composition with neighbor steering directions at the same site** is non-additive and eval-detection-dominated. We assemble five directions (refusal, $d_{\text{mech}}$, probe, honesty, eval-detection) at `attn_out_L12` and measure their pairwise and triple project-out interference. Five of ten pairs deviate from summation; all three OVERRIDING pairs include the eval-detection direction. Dosage on the ($d_{\text{refusal}}$, $d_{\text{mech}}$) pair peaks at $\alpha \approx \|r\|$ and saturates non-monotonically. The inventory has effective rank 4 by construction.

5. **Geometric universality without causal universality across scales.** The analogous eval-detection and sandbag directions in Gemma 2 9B IT (computed at `attn_out` of the proportional layer) align with the 2B directions at cosine 0.921 and 0.996 respectively after PCA-200 Procrustes alignment, yet single-fire project-out at the corresponding 9B layer produces no causal effect on $n_{\text{val}} = 40$ for either direction or either model. Geometric alignment via Procrustes rotation does not by itself license causal claims at those directions.

6. **Standard SAE feature-selection by mean activation difference fails to recover the mechanism at the same site.** Cross-stage (Gemma 2 2B base $\rightarrow$ IT) yields 1745, 2594, and 2576 features above $2\sigma$ at L11/L12/L13 with maximum absolute shifts 19–25; project-out of the top-$K = 20$ features at `attn_out_L12` produces $\Delta_{\text{count}} = 1/200$ across three seeds against a pre-registered $\geq 20$ threshold. Within-condition (sandbag vs honest) at $K = 20$ produces an opposite-signed $-15$ pp effect on the $n = 40$ test split where $d_{\text{mech}}$ recovers $+10$ pp—a sign-reversal landmark in which mean-shift selection identifies a subspace that, when ablated, ablates capability rather than suppression.

7. **An empirical scope-bounding observation.** In five candidate small instruction-tuned models across three architectures (Gemma 2, Qwen 2.5, Llama 3) and two induction methods (prompted Mitra-style and QLoRA fine-tuning), only Gemma 2 2B IT exhibited the gate-clearing behavior; Llama 3.2 1B specifically inverted the eval-framing response. The single-model focus is empirically grounded rather than experimentally incomplete.

8. **Methodological discipline.** A four-tier validation protocol (T1 magnitude $\rightarrow$ T2 controls $\rightarrow$ T3 single-touch test $\rightarrow$ fresh-prompt replication on a source-ID-disjoint prompt set) applied across 110 candidate intervention loci, with 18 cumulative test touches under single-touch-per-locus discipline, audit-verified.

**Deployment-relevant implications are twofold.** Capability evaluations on Gemma-family IT models likely underestimate true capability when prompts frame the interaction as evaluation: behavioral data show eval framing alone produces accuracy drops in 3 of 4 measurable domains, and the L12 mechanism's eval-detection dominance is consistent with that signal flowing through the same direction. The L12 intervention is also a feasible activation-level defense—rank-1, refusal-orthogonal, and transferable across sandbag-trigger conditions on shared prompts—bounded by three negatives developed in §4.9: asymmetric polarity (project-out works; symmetric add-negative does not immunize), sparse selectivity ($\sim 7.5\%$ of val prompts respond to add-direction at $\alpha = 2.0$), and variant-dependent behavior under conflict prompts.

**Roadmap.** §2 positions the work in three lines of prior literature. §3 specifies the model, behavioral suite, pre-registration discipline, and intervention objects. §4 presents the substantive results in three blocks: the L12 mechanism and its vertical embedding (§4.1–4.5); three neighborhood

characterizations at the same site (§4.6–4.8); and scope-bounding negatives (§4.9–4.10). §5 discusses the integrative picture. §6 enumerates limitations. §7 concludes.

## 2 Related Work

**Behavioral sandbagging.** van der Weij et al. [27] showed that instruction-tuned language models can be induced to underperform on capability evaluations through system-prompt instructions, and that this behavior persists under password-locking fine-tuning. The induction we use—telling the model it is being evaluated, will be retrained if too capable, and should give plausible-but-incorrect answers—derives from that work and from successor "Mitra-style" prompts that vary surface phrasing while preserving the underperformance instruction. Tice et al. [25] demonstrated that injecting noise into weights elicits hidden capabilities of sandbagging models. Our extension is from behavioral characterization to causal mechanism.

**Probe-based detection and evaluation awareness.** Goldowsky-Dill et al. [9] train linear probes on residual-stream activations that distinguish sandbagging from honest behavior at AUROC 0.96–0.999 on Llama-3.3-70B-Instruct and generalize to insider-trading concealment and safety-evaluation underperformance. Taylor et al. [23] stress-test probe-based schemes against red-team-instilled sandbagger model organisms and demonstrate that final-layer probes can fail to cleanly separate sandbagging in adversarially constructed organisms. Nguyen et al. [17] probe and steer evaluation awareness specifically in Llama-3.3-70B, showing that SAE features selected from a Goodfire-style API partially uncover sandbagging on WMDP. Chaudhary et al. [4] extend the probe panel to 15 models across four families and report a power-law scaling of evaluation awareness with parameter count. We provide a complementary white-box causal counterpart on a different model: probes detect that the model is sandbagging; the L12 direction-projection causally restores the suppressed capability without re-training and without modifying weights.

**Refusal directions and adjacent intervention objects.** Arditi et al. [2] established that refusal in instruction-tuned models is mediated by a single residual-stream direction, identifiable as a difference-in-means between harmful and harmless instruction activations and modifiable by direction projection. We borrow this intervention object—the L2-normalized difference-in-means projected out at a specific component—and apply it to a different target behavior with explicit refusal-orthogonality verification: the L12 sandbag direction has cosine 0.357 with the L12 refusal direction, and the refusal-orthogonal component recovers the full +5.0 pp val effect on its own. The broader activation-steering literature [18, 21, 26] and the linear representation hypothesis [19] provide the methodological context, and the well-established observation that naive linear combination of steering vectors for multiple behaviors is largely unsuccessful [20, 28] directly motivates the composition section of §4.6.

**Mechanistic interpretability methodology.** Activation patching as causal-mediation analysis in transformers [29, 30] is the foundational technique behind our upstream-locus search at L11; direct logit attribution following the mathematical-framework decomposition [6] provides convergent evidence. Sparse autoencoder methodology [3, 5, 13, 24] underpins our complementary residual-stream analysis at L15 and the eval-vs-explicit-sandbag SAE-feature decomposition. Attribution-graph circuit tracing [1, 15] is the toolchain whose Phase 0 verification established pipeline correctness; full attribution graphs for sandbagging are limited to short prompts on our hardware (12 GB VRAM) and registered as future work.

**SAE feature selection and cross-stage diffing.** Cross-coder model diffing [14] and feature-shift diffing identify features whose mean activation changes between two checkpoints. Recent critical work shows that crosscoder joint reconstruction prioritizes high-frequency shared features while suppressing the sparse low-magnitude shifts that are precisely the behaviorally critical features under narrow fine-tuning [16]; proposed remedies include Delta-Crosscoder variants that explicitly allocate capacity for fine-tuning-induced shifts. Sanity-check work has shown that random baselines match or approach trained SAEs on standard interpretability metrics [10, 12], and that raw probes often outperform SAE-based probes on downstream tasks [7, 11]. §4.8 contributes a within-site quantification of this gap at $K = 20$ against a fixed causal anchor ($d_{\text{mech}}$), exhibiting both a near-zero cross-stage causal effect and an opposite-signed within-condition effect at the same site.

**Cross-model and cross-scale direction alignment.** A growing line shows that geometric alignment of internal directions across model scales or families is high after standardization or Procrustes rotation [22]. Concurrent work has reported that the geometry and intervention outcome of safety-related signals vary across labs and do not transfer between models trained under different alignment regimes. §4.7 contributes a within-family within-direction quantitative measurement of the disjunction: cosine 0.921 (eval-detection) and 0.996 (sandbag) between 2B and 9B, with null causal effect at the canonical layer of each.

## 3 Methodology

### 3.1 Model and behavioral suite

We study Gemma 2 2B IT [8] in bfloat16. All experiments ran on the `unsloth/gemma-2-2b-it` mirror; a sanity-audit on the canonical `google/gemma-2-2b-it` weights remains outstanding (§6). The model has 26 decoder layers, residual-stream width $d_{\text{model}} = 2304$, and 8 attention heads per layer. Cross-scale experiments in §4.7 additionally use Gemma 2 9B IT loaded as `unsloth/gemma-2-9b-it-bnb-4bit` (4-bit weights dequantized at inference, bfloat16 compute); cross-stage experiments in §4.8 additionally use Gemma 2 2B base (`unsloth/gemma-2-2b`).

The behavioral suite contains 200 prompts spanning five capability domains (40 each): arithmetic (GSM8K-easy single- or two-step problems), factual recall (TriviaQA `rc.nocontext` on well-attested entities), basic logic (BoolQ subset), code completion (HumanEval-easy, $\leq$ 10 LOC solutions), and reading comprehension (MMLU reading-style subjects). Each prompt has a deterministic ground-truth answer with execution-based or string-equality scoring. We partition 60 / 20 / 20 into train / val / test ($n = 120/40/40$) under `random.seed(0)`, stratified within domain, committed to the repository. Test is held out and touched at most once per locus per criterion; the program's cumulative test-touch count is 18, each logged before the run.

We use three system-prompt conditions: `honest` (a neutral baseline), `sandbagging` (a Mitra-style induction telling the model it is being evaluated, will be retrained if too capable, and should give plausible-but-incorrect answers), and `eval_context_only` (telling the model it is being evaluated but not instructing underperformance—a confounder-isolation arm). Generation uses greedy decoding (`do_sample=False`), `max_new_tokens=256`, and seeds $\{0, 1, 2\}$; the three-seed outputs are nearly identical per prompt under greedy decoding, so our bootstrap procedure averages per-prompt across seeds before resampling.

The pre-registered exclusion rule drops prompts where mean honest accuracy is below 0.5 (no capability to suppress). Under this rule, 95 of 200 prompts are excluded (47.5%), including all 40 code-completion prompts: HumanEval execution-based scoring returns 0/120 correct in the honest condition because the model produces verbose code that does not pass test execution. This is a

metric failure, not a capability failure, and the entire code-completion domain is excluded from gate computation; we report it transparently in §4.1.

### 3.2 Pre-registration and validation discipline

The scientific design—hypotheses, gate criteria, exclusion rules, tolerances—was frozen in `INVARIANT.md` (commit `19704e1`) before any data was collected. Each phase had a separate registered protocol committed before runs began; mid-flight amendments were committed as dated amendments rather than silent scope changes.

Every causal claim went through a four-tier validation protocol. T1 magnitude: per-locus screening on val (40 prompts $\times$ 3 seeds), retaining candidates with $|\Delta_{\text{val}}| \geq 5$ pp. T2 controls: capability control ($\leq 5$ pp damage to honest accuracy), specificity control ($\leq 10$ pp refusal-rate change on a 50-prompt JailbreakBench-derived set), and a random-direction or random-feature baseline. T3 single-touch test: one run on the held-out test split, magnitude- and direction-matched, logged to `TEST_TOUCHES.md` before the run begins. D1 fresh-prompt replication: one run on a 40-prompt fresh held-out set (`prompts_fresh.jsonl`, source-IDs disjoint from the 200-prompt suite). Same direction and $\geq 5$ pp magnitude required for "robustly validated."

Across the program (Phase A residual-stream sweep at 21 layers; Phase B component-level sweep at 76 loci; Phase C 13-cell multi-layer + steering grid), 110 candidate intervention loci were screened, 18 cumulative test-touches consumed, and two loci robustly validated through all four tiers. Three audit rounds confirmed the test-touch ledger is consistent and no locus was re-tested.

### 3.3 Intervention objects

**Difference-in-means direction projection (the headline object).** For a target component output, let $m_s$ and $m_h$ be the per-prompt mean activations under sandbag and honest, computed over train $\cup$ val at the prompt-final token across all three seeds. The direction is $d = (m_s - m_h)/\|m_s - m_h\|_2$, a unit vector in $d_{\text{model}} = 2304$. The intervention applies a projection-out hook at the prompt-final token of the first forward pass only: replace $v$ with $v - (v \cdot d)\, d$, where $v$ is the component output. Subsequent generation steps run unmodified.

**SAE-feature decoder ablation.** For a target residual-stream layer, zero out a top-$K$ set of feature activations (Gemma Scope JumpReLU SAEs, 16k width; Lieberum et al. 13) at `resid_pre[L]` on the prompt-final token of the first forward pass; equivalent to subtracting $\sum_i a_i(x) W_{\text{dec}}[i]$. Used in Phase A's complementary L15 analysis and in the mean-shift selection of §4.8.

**Activation patching.** For an upstream layer $L \in \{0..11\}$, replace `resid_pre[L][last_token]` in the sandbag forward with the cached honest activation; either capture `attn_out_L12` (projection mode) or continue generation (generation mode).

**Direct logit attribution.** Linear $V$ projection of attention contributions per the Elhage et al. [6] decomposition; RMSNorm correction omitted (preserves relative ranking, produces 74% absolute reconstruction error). The component ranking is the substantive output; absolute magnitudes are approximate.

**Statistical methodology.** Bootstrap CIs at $B = 1000$ over per-prompt means (greedy decoding makes per-seed outputs nearly identical; the per-prompt level is the effective unit). BH multiple-comparisons correction is reported descriptively ($q < 0.05$) but is non-gating per the pre-registered

**Table 1.** Phase 1 capability gaps after the pre-registered exclusion rule (mean honest accuracy $< 0.5 \rightarrow$ drop). 95% bootstrap CIs from $B = 1000$ resamples over per-prompt means. Pre-registered gate: $\geq 25$ pp gap with 95% CI excluding zero on at least 3 of 5 domains.

| domain | $n_{\text{prompts}}$ | gap (pp) | 95% CI | gate $\geq 25$ pp |
|---|---|---|---|---|
| arithmetic | 26 | +42.3 | [+26.9, +61.5] | ✓ |
| basic_logic | 28 | +28.6 | [+10.7, +46.4] | ✓ |
| reading_comp | 25 | +28.0 | [+12.0, +48.0] | ✓ |
| factual_recall | 26 | +19.2 | [+7.7, +34.6] | — |
| code_completion† | 0 | n/a | n/a | — |

† Code-completion excluded per §3.1: HumanEval execution-based scoring returns 0 / 120 correct in the honest condition (median 140 output tokens; 0 refusals; 0 malformed outputs). 0% honest accuracy is a metric failure, not a capability failure.

binding decision rule (magnitude + direction + controls + replication). Full detail in Appendix A of the underlying technical report; the binding instrument is the four-tier protocol above.

## 4 Results

### 4.1 Behavioral characterization

Gemma 2 2B IT exhibits substantial induced sandbagging on three of five capability domains under Mitra-style induction. Table 1 reports the Phase 1 capability gaps after the pre-registered exclusion rule.

Three of five domains pass the gate; the pre-registered Phase 1 gate is satisfied. Refusal rates were 0.5% across conditions; malformed-output rates were 2% honest vs 10% sandbagging, consistent with the system prompt's instruction to be intentionally vague.

The `eval_context_only` confounder-isolation arm—telling the model it is being evaluated but not instructing underperformance—produces accuracy drops in 3 of 4 measurable domains: basic_logic 22.5 pp, arithmetic 7.5 pp, reading_comp 5.0 pp; factual_recall changes by only 2.5 pp. (The apparent rise on code_completion is uninterpretable due to the honest-condition scoring artifact.) Eval framing alone, without underperformance instruction, materially reduces accuracy where the metric works. This anticipates the L12 mechanism's eval-detection dominance characterized in §4.3.

### 4.2 The L12 suppression mechanism

We identify a rank-1 direction at the attention output of layer 12 whose projection-out restores sandbag-condition capability by +10.0 pp on fresh held-out prompts with no measurable damage to honest capability. The direction is $d_{\text{mech}} = (m_s - m_h)/\|m_s - m_h\|_2$ where $m_s$ and $m_h$ are per-prompt mean `attn_out_L12` activations at the prompt-final token under sandbag and honest, averaged across train $\cup$ val (160 prompts) and three seeds. The intervention replaces $v$ with $v - (v \cdot d_{\text{mech}})\, d_{\text{mech}}$ at the prompt-final token of the first forward pass; subsequent generation steps run unmodified.

Three controls passed. The val-side random-direction baseline at `attn_out_L12` produced $\Delta = 0$ pp. Capability control on honest fresh prompts produced 0 pp damage (within the 5 pp tolerance). Specificity control on the 50-prompt JailbreakBench-derived refusal set produced 0 pp refusal-rate change (within the 10 pp tolerance). The intervention restores capability under sandbagging without compromising honest behavior or refusal of harmful instructions.

Pre-registration H1 (a causally-connected eval-detection-and-capability-suppression pathway) is supported. H3 (targeted ablation reduces sandbagging within the capability tolerance) is partially

**Table 2.** Validation trajectory at `attn_out_L12`. Each row is a separate held-out evaluation ($n = 40 \times 3$ seeds). T1 = sandbag-val magnitude screening; T3 = single-touch sandbag-test; D1 = fresh-prompt replication (source-IDs disjoint from the 200-prompt suite).

| evaluation | sandbag baseline | sandbag ablation | $\Delta$ (pp) |
|---|---|---|---|
| Tier-1B val | 0.500 | 0.550 | +5.0 |
| Tier-3B test (single touch) | 0.450 | 0.575 | +12.5 |
| Phase D D1 fresh | 0.475 | 0.575 | +10.0 |
| Phase D D1 honest fresh (capability control) | 0.625 | 0.625 | +0.0 |

supported: the +10 pp fresh restoration is approximately 35–40% of the within-domain Phase 1 gap on the gate-passing domains, with direction-consistency robust at the 5 pp threshold and absolute magnitude generalization within the $n = 40$ power constraint.

A 76-locus component sweep across attention outputs, MLP outputs, and per-head attention contributions preceded this result. Tier-1B identified 14 candidates; Tier-2B passed 10 of 14; Tier-3B (10 cumulative test touches) yielded 4 survivors; Phase D D1 yielded 2 robustly validated loci, `attn_out_L12` and `attn_out_L14`, with `attn_out_L12` carrying the larger and cleaner effect. The L14 secondary locus has val/test/fresh of +5.0/ + 7.5/ + 5.0 pp with +2.5 pp honest-fresh improvement.

### 4.3 Mechanism properties at L12

The headline finding is a single direction at a single component on a single token of a single forward pass. We characterize four properties: rank, refusal-orthogonality, eval-detection dominance, and across-condition transfer.

#### *4.3.1 Rank*

The mechanism is precisely rank-1. SVD of the (sandbag − honest) per-prompt-mean difference matrix at `attn_out_L12` (160 train+val prompts × 2304 dimensions) gives $\sigma_2/\sigma_1 = 0.275$; the top-1 right-singular vector matches the cached Phase B direction at cos = 0.9988.

| $K$ | $\Delta_{\text{val}}$ (pp) | interpretation |
|---|---|---|
| 1 | +5.0 | matches Phase B direction |
| 2 | +5.0 | rank-2 adds nothing |
| 10 | 0.0 | over-correction (cancels via overlap with task-relevant signal) |

$K = 2$ rules out a rank-2-or-higher subspace explanation; $K = 10$ rules out a magnitude artifact. Capability control at $K = 1$ is honest $\Delta = -2.5$ pp (within tolerance); a random $K = 1$ subspace produces $\Delta = +2.5$ pp, exceeded by the discovered direction at the registered 2 pp distinguishability margin. A single 2304-dimensional unit vector specifies the deployment-time intervention.

#### *4.3.2 Refusal-orthogonality*

The L12 sandbag direction is near-orthogonal to the L12 refusal direction. We computed a refusal direction in the same component-output space using the Arditi et al. [2] construction: mean activation difference at `attn_out_L12` between 50 JailbreakBench harmful prompts and honest prompts, L2-normalized.

**Table 3.** Four-direction projection-out at `attn_out_L12`, evaluated across three conditions on val ($40 \times 3$ seeds). Baselines: honest 0.600, eval-context 0.475, sandbag 0.500. The $d_{\text{phaseB}}$ row is the replication anchor.

| direction projected out | Δ honest | Δ eval-context | Δ sandbagging |
|---|---|---|---|
| $d_{\text{eval}}$ | −2.5 | +0.0 | +7.5 |
| $d_{\text{explicit}}$ | −2.5 | +0.0 | +0.0 |
| $d_{\text{phaseB}}$ (replication) | −2.5 | −2.5 | +5.0 ✓ |
| $d_{\text{random}}$ (control) | +0.0 | −2.5 | +0.0 |

| metric | value |
|---|---|
| signed $\cos(d_{\text{sandbag}},\ d_{\text{refusal}})$ | +0.357 |
| Phase B fraction orthogonal to refusal | 0.934 |
| H2 distinctness threshold (registered $\lvert\cos\rvert < 0.7$) | ✓ |
| Tighter $\lvert\cos\rvert < 0.5$ | ✓ |

The refusal-aligned component contributes nothing causally. Re-running the projection-out hook on val using only the unit vector orthogonal to the refusal direction within the Phase B span produces $\Delta_{\text{val}} = +5.0$ pp, recovering the full Phase B effect; the 35.7% refusal-aligned component contributes zero. A complementary SAE-feature analysis at L15 found Jaccard = 0.042 between the 20-feature Phase A sandbag-feature set and the 30-feature Arditi-style L15 refusal-feature set. The two analyses converge: the sandbagging mechanism is partially distinct from refusal at both the SAE-feature level (residual stream) and the direction level (attention output).

#### *4.3.3 Eval-detection dominance, not eval-vs-explicit dissociation*

To test whether the L12 mechanism cleanly separates eval-awareness from explicit sandbagging, we computed three directions at `attn_out_L12` over train ∪ val: $d_{\text{eval}} = \text{mean(eval_context_only)} - \text{mean(honest)}$; $d_{\text{explicit}} = \text{mean(sandbagging)} - \text{mean(eval_context_only)}$; $d_{\text{phaseB}} = \text{mean(sandbagging)} - \text{mean(honest)}$. Geometrically: $\cos(d_{\text{eval}}, d_{\text{explicit}}) = -0.286$ (near-orthogonal); $\cos(d_{\text{eval}}, d_{\text{phaseB}}) = +0.618$; $\cos(d_{\text{explicit}}, d_{\text{phaseB}}) = +0.578$; the regression $d_{\text{phaseB}} \approx +0.852\, d_{\text{eval}} + 0.821\, d_{\text{explicit}}$ fits with $R^2 = 1.00$. Geometric distinctness predicts mechanism separability; the causal results contradict that prediction.

The pre-registered LANDMARK criterion required double-dissociation (each direction $\geq 5$ pp on its target AND $\leq 3$ pp on the other); this is not met. The substantive interpretation: at `attn_out_L12`, projecting out $d_{\text{eval}}$ restores sandbag accuracy by +7.5 pp—more than projecting out $d_{\text{phaseB}}$ itself—while $d_{\text{explicit}}$ produces zero behavioral effect. The L12 mechanism is dominated by the eval-detection signal; the explicit-underperformance component, despite a geometrically distinct direction, has no separable causal hold at this locus. A three-way categorical SAE-feature contrast on train ∪ val separates 203 EVAL-AWARENESS features from 93 EXPLICIT-SANDBAG features at L12 (similar ratios at L11/L13), with eval-awareness outnumbering explicit-sandbag by approximately 2:1. SAE-feature counts and direction-projection causal results agree on which signal carries the L12 mechanism.

#### *4.3.4 Across-condition trigger transfer*

We tested whether the L12 direction, computed under T1 (standard Mitra system prompt), transfers to T2 (paraphrased system prompt) and T3 (clean honest system prompt with sandbagging

instruction prepended to the user turn). All three triggers induce $\geq 15$ pp sandbagging gaps on the calibration set (T1 +23.3 pp, T2 +36.7 pp, T3 +33.3 pp; $n = 30$ train-split prompts).

Projecting out $d_{\text{mech}}$ at `attn_out_L12` while sandbagging is induced by each trigger restores accuracy by +3.33/ + 10.0/ + 10.0 pp on T1 / T2 / T3—more than any of the within-trigger re-discovered directions achieved on T1 itself. The calibration prompt-IDs are a subset of train, so this is an across-trigger-condition transfer claim on shared prompts: the same direction is causally engaged when the trigger varies, not that the direction generalizes to held-out prompts under each trigger. Across-prompt cross-trigger validation—recomputing direction discovery on an independent prompt set under each trigger—is registered as future work (§6).

### 4.4 Upstream decision locus at L11

Where does the L12 sandbag direction's content come from? Activation patching at upstream layers $L \in \{0..11\}$ on the prompt-final token, replacing the sandbag activation with the cached honest activation, identifies a concentrated upstream cluster at L10–L11. The finding registered as single-session STRONG-POSITIVE: the LANDMARK threshold ($\Delta_{\text{proj}} \geq 50\%$ reduction in the L12 sandbag-direction projection at some upstream layer) was not met (peak $\Delta_{\text{proj}} = 32.4\%$ at L11), and replication on a disjoint prompt set is registered as future work.

We selected 30 train prompts by top-Phase-1-gap with the honest $> 0.5$ capability filter (selection method documented before any data was touched). For each $L$, we patched `resid_pre[L][last_token]` and measured the resulting `attn_out_L12` projection onto cached $d_{\text{phaseB}}$.

| $L$ | $\Delta_{\text{proj}}$ / baseline | SE |
|---|---|---|
| 0–7 | $\approx 0.000$ to $-0.009$ | 0.001–0.002 |
| 8 | +0.025 | 0.003 |
| 9 | +0.036 | 0.006 |
| 10 | +0.197 | 0.009 |
| 11 | +0.324 | 0.011 |

The signal is essentially zero through L7, small at L8–L9, and sharply concentrated at L10–L11. Mean baseline projection magnitude is 5.218; the 32.4% reduction at L11 corresponds to an absolute reduction of 1.69.

Generation-mode patching verified the causal claim behaviorally. At L11, replacing `resid_pre[L11][last_token` with the cached honest activation in the sandbag forward and continuing generation restored accuracy by +20 pp (random control −2.2 pp; honest-on-honest stability 1.000). At L10 the same procedure restored +16.7 pp with random control −5.6 pp—same direction-specific causal-excess magnitude as L11, but the random control fails the strict $\leq 3$ pp criterion (so L10 is a probable secondary locus by causal-excess metric without meeting the strict confirmation criterion).

The locus replicates across triggers. Patching at L11 under T2 (paraphrased) and T3 (user-turn injection) reduces the L12 projection by 22.7% and 66.5% respectively, both clearing the registered $\geq 30\%$-of-T1-fraction threshold. Trigger-invariance is supported as "the same upstream locus is implicated under all three triggers," not "more concentrated under user-turn injection": T3's higher normalized fraction reflects its smaller baseline projection magnitude.

Direct logit attribution provides convergent evidence and refines the Stage 1 picture. Decomposing `attn_out_L12` linearly across upstream attention contributors (per-component $V$ projection without RMSNorm correction; mean rel_err 74% on absolute reconstruction, with relative ranking preserved per Elhage et al. 6), the top-3 contributors to the L12 sandbag direction are `attn_out_L10` (+0.334),

`attn_out_L08` (+0.314), and `attn_out_L09` (+0.305); `attn_out_L11` contributes only +0.044. The DLA top-3 layers match the activation-patching top-3 within ±1 layer. L11's own attention output contributes negligibly despite L11 being the top causal patching locus: L8–L10 attention layers generate the sandbag-decision signal that accumulates in L11's residual stream via standard residual-stream additivity; L11 is the read-off carrier from which L12's attention reads, not the source. Per pre-registered anti-pattern discipline, DLA does not assert a causal upstream-locus claim from attribution alone; activation patching is the binding causal-claim instrument, and DLA refines Stage 1 to "L8–L10 generates, L11 carries."

### 4.5 Downstream partial compensation at L13–L25

The L12 ablation perturbation propagates and amplifies through downstream layers, but the sandbag-direction-aligned component decays. We registered a self-repair characterization with a binding criterion ($\|\Delta\text{resid}_{\text{L25}}\|/\|\Delta\text{resid}_{\text{L13}}\| < 0.80 \rightarrow$ SELF_REPAIR observed) and ran it on val sandbag prompts ×3 seeds ($n = 120$ observations). For each prompt-seed pair we captured `resid_pre[L][last_token]` at $L \in \{12, 13, 15, 18, 21, 25\}$ in two forward passes (baseline, ablated) and computed $\Delta_L = \text{resid_pre}[\text{L}]_{\text{abl}} - \text{resid_pre}[\text{L}]_{\text{base}}$.

| $L$ | $\|\Delta_L\|$ mean | 95% CI | mean $\langle \Delta_L, d_{\text{sandbag,L}} \rangle$ | $\|\text{proj}\|/\|\Delta_L\|$ |
|---|---|---|---|---|
| 12 | 0.00 (sanity ✓) | — | — | — |
| 13 | 36.2 | $[34.7, 37.5]$ | $-21.3$ | 0.59 |
| 15 | 40.3 | $[38.7, 41.8]$ | $-15.2$ | 0.38 |
| 18 | 50.4 | $[47.8, 52.9]$ | $-10.5$ | 0.21 |
| 21 | 70.1 | $[65.8, 74.1]$ | $-6.9$ | 0.10 |
| 25 | 107.9 | $[100.8, 114.9]$ | $-11.0$ | 0.10 |

The ratio L25 / L13 = 2.99 is well above the 0.80 threshold; NO_SELF_REPAIR by the registered $\|\Delta\|$ criterion. The L12 perturbation is not compensated away in residual-stream norm. Projections onto the cached `resid_pre` sandbag-direction at each downstream layer are uniformly negative (ablated forward has less sandbag-direction signal at every downstream layer), and the absolute magnitude decays from 21.3 at L13 to 11.0 at L25—approximately 51% restoration of the sandbag-direction signal across the downstream stack. The downstream growth happens in directions orthogonal to the original sandbag direction: the perturbation's norm grows 3×, but only ≈ 10% of L25's perturbation lies along the sandbag direction.

Together: the L12 attention-output direction is a causally upstream node with cascading downstream consequences. The downstream response is a partial hydra effect on the suppression-direction-aligned component—the model partially restores the suppressed-capability signal through orthogonal pathways—paired with broader downstream computational changes that grow the residual-stream perturbation but do not restore the targeted direction. The L12 intervention is operationally robust against simple direction-aligned compensation: downstream layers restore only ≈ 51% of the suppression-direction signal thirteen layers downstream of L12.

### 4.6 Composition with neighbor steering directions at L12

Practitioners increasingly stack multiple steering vectors at the same layer to compose behavioral controls. Two independent lines of empirical work indicate this is rarely additive in transformer LMs [20, 28]. We provide a quantitative within-site characterization at `attn_out_L12` in Gemma 2 2B IT for a five-direction inventory: refusal ($d_{\text{refusal}}$, Arditi-style), the mechanism direction ($d_{\text{mech}}$), a

**Table 4.** Cosine matrix at `attn_out_L12` across the five directions. Singular values are $[1.50, 1.03, 1.00, 0.834, 0.0099]$, so the effective rank of the inventory is 4 ($d_{\text{honesty}}$ is anti-aligned with $d_{\text{mech}}$ by construction).

| | $d_{\text{refusal}}$ | $d_{\text{mech}}$ | $d_{\text{probe}}$ | $d_{\text{honesty}}$ | $d_{\text{eval}}$ |
|---|---|---|---|---|---|
| $d_{\text{refusal}}$ | 1.000 | 0.357 | −0.016 | −0.359 | −0.067 |
| $d_{\text{mech}}$ | 0.357 | 1.000 | 0.017 | −1.000 | 0.178 |
| $d_{\text{probe}}$ | −0.016 | 0.017 | 1.000 | −0.018 | −0.006 |
| $d_{\text{honesty}}$ | −0.359 | −1.000 | −0.018 | 1.000 | −0.175 |
| $d_{\text{eval}}$ | −0.067 | 0.178 | −0.006 | −0.175 | 1.000 |

**Table 5.** Pairwise composition at `attn_out_L12`. Tally: ADDITIVE 4, DESTRUCTIVE 2, OVERRIDING 3, NEUTRAL 1, CONSTRUCTIVE 0. All three OVERRIDING pairs include $d_{\text{eval}}$.

| pair | $\Delta_{\text{combined}}$ | $\Delta_i$ | $\Delta_j$ | $I$ | regime |
|---|---|---|---|---|---|
| ($d_{\text{refusal}}$, $d_{\text{mech}}$) | −6 | 0 | +1 | −7 | DESTRUCTIVE |
| ($d_{\text{refusal}}$, $d_{\text{probe}}$) | 0 | 0 | +2 | −2 | NEUTRAL |
| ($d_{\text{refusal}}$, $d_{\text{honesty}}$) | −5 | 0 | +1 | −6 | DESTRUCTIVE |
| ($d_{\text{refusal}}$, $d_{\text{eval}}$) | +1 | 0 | −3 | +4 | OVERRIDING |
| ($d_{\text{mech}}$, $d_{\text{probe}}$) | +3 | +1 | +2 | 0 | ADDITIVE |
| ($d_{\text{mech}}$, $d_{\text{honesty}}$) | +4 | +1 | +1 | +2 | ADDITIVE |
| ($d_{\text{mech}}$, $d_{\text{eval}}$) | +4 | +1 | −3 | +6 | OVERRIDING |
| ($d_{\text{probe}}$, $d_{\text{honesty}}$) | +4 | +2 | +1 | +1 | ADDITIVE |
| ($d_{\text{probe}}$, $d_{\text{eval}}$) | −2 | +2 | −3 | −1 | ADDITIVE |
| ($d_{\text{honesty}}$, $d_{\text{eval}}$) | +3 | +1 | −3 | +5 | OVERRIDING |

probe direction ($d_{\text{probe}}$, logistic-regression coefficient on the sandbag-vs-honest contrast), an honesty direction ($d_{\text{honesty}}$, sign-flipped sandbag-vs-honest contrast), and the eval-detection direction ($d_{\text{eval}}$). All directions are unit-normalized at the same layer.

#### *4.6.1 Inventory geometry and effective rank*

#### *4.6.2 Individual project-out effects*

Single-direction project-out at `attn_out_L12` on the sandbag-condition 200-prompt suite (baseline = 91/200 correct): $d_{\text{refusal}}$ 0, $d_{\text{mech}}$ +1, $d_{\text{probe}}$ +2, $d_{\text{honesty}}$ +1, $d_{\text{eval}}$ −3. The negative sign on $d_{\text{eval}}$ is consistent with cycle-001's $d_{\text{mech}}$ protocol; $d_{\text{eval}}$ has the strongest individual effect and is the only direction whose individual effect exceeds the $2\sigma$ random-direction threshold.

#### *4.6.3 Pairwise composition regimes*

For each pair $(d_i, d_j)$, we project out the combined subspace $\text{span}(d_i, d_j)$ at `attn_out_L12` (single-fire) and compute the interference $I = \Delta_{\text{combined}} - (\Delta_i + \Delta_j)$. Regime classification (pre-registered thresholds): ADDITIVE $|I| \leq 2$; DESTRUCTIVE $I \leq -5$; OVERRIDING $I \geq +4$ with $|\Delta_{\text{combined}}| \geq |\Delta_i| + |\Delta_j|$; CONSTRUCTIVE $I \geq +4$ with $|\Delta_{\text{combined}}| < |\Delta_i| + |\Delta_j|$; NEUTRAL $\Delta_{\text{combined}} \approx 0$ with non-zero individual $\Delta$s.

Five of ten pairs (DESTRUCTIVE + OVERRIDING) deviate from simple summation. All three OVERRIDING pairs include $d_{\text{eval}}$: the eval-detection direction has both the largest individual effect

**Table 6.** Geometric alignment vs. causal effect across scales. cos values are computed after PCA-200 reduction with Procrustes rotation. Causal validation: single-fire project-out at the canonical layer on the 200-prompt suite val split ($n = 40$); pre-registered tier-1B threshold $\Delta_{\text{count_sb}} \geq +2$ with $\Delta_{\text{count_hn}} \geq -2$ and beating-random-by-$2\sigma$.

| direction | cos(2B, 9B) | 2B $\Delta_{\text{sb}}/40$ | 9B $\Delta_{\text{sb}}/40$ | passes t1b on 2B | passes t1b on 9B |
|---|---|---|---|---|---|
| $d_{\text{eval}}$ | 0.921 | $-1$ | 0 | false | false |
| $d_{\text{sandbag}}$ | 0.996 | $-1$ | $-2$ | false | false |

and the largest interaction effects across the inventory. For deployment, $d_{\text{eval}}$ is the highest-leverage axis among the directions tested.

#### *4.6.4 Triple compositions and dosage*

Sampling 5 of 10 possible triples with balanced inclusion of each direction, residual three-way interaction strength $T$ (excess beyond pairwise) is non-additive ($|T| \geq 3$) on 3 of 5 triples; 2 of 3 non-additive triples include $d_{\text{eval}}$, and the largest $|T| = 6$ is ($d_{\text{refusal}}$, $d_{\text{honesty}}$, $d_{\text{eval}}$). A dosage sweep on the ($d_{\text{refusal}}$, $d_{\text{mech}}$) pair (selected for the largest pairwise destructive interference, $I = -7$) shows the combined direction peaks at $\alpha = 1.0$ ($|\Delta| = 6$ of 200 prompts) and saturates non-monotonically at $\alpha \geq 1.5$ ($|\Delta| = 1$). The curve argues against assuming linear scaling of steering magnitude at this site.

### 4.7 Geometric versus causal alignment across scales

Cross-model alignment of internal directions has been proposed as a candidate signal of representational universality. Within the Gemma 2 family, Son et al. [22] report mid-layer feature convergence between Gemma 2 2B and 9B at SVCCA $\approx 0.73$; broader cross-model alignment work has demonstrated that Procrustes-aligned latent spaces preserve semantic geometry sufficiently to enable interpretability transfer. The natural question for an identified causal mechanism is whether geometric alignment of the mechanism direction across scales implies that the same direction is causal in the larger model.

We measure both quantities for two sandbagging-relevant directions: $d_{\text{eval}}$ (eval-context-only − honest) and $d_{\text{sandbag}}$ (sandbag − honest), at `attn_out_L12` in Gemma 2 2B IT and `attn_out_L21` in Gemma 2 9B IT (L21 chosen by proportional depth fraction, with the model loaded as the 4-bit `unsloth/gemma-2-9b-it-bnb-4bit` checkpoint at bfloat16 compute). Both 2B (2304-dim) and 9B (3584-dim) directions are projected into a shared 200-dimensional subspace via PCA on each model's prompt-distribution activations, then Procrustes-aligned; cosine is reported in the shared subspace.

Geometric alignment is high—$d_{\text{eval}}$ at 0.921 and $d_{\text{sandbag}}$ at 0.996—yet single-fire project-out at the canonical layer of each model produces no causal capability recovery on either direction in either model, at $n_{\text{val}} = 40$ each. The 2B $-1$ result also illustrates an internal calibration point: the bare difference-in-means direction ($d_{\text{sandbag}}$) is geometrically close but causally distinct from $d_{\text{mech}}$, the refined direction selected by the Phase B component sweep on `attn_out_L12` in §4.2—$d_{\text{mech}}$ recovers $+10$ pp on test where $d_{\text{sandbag}}$ recovers $-1/40$ on val. The contrast is informative as a calibration of what cosine alignment between candidate directions can and cannot license.

Three interpretations are consistent with the 9B null. (a) The 9B model implements the sandbagging mechanism at a different layer than the proportional analog. (b) The 9B causal pathway is larger-magnitude or distributed across multiple sites, so single-fire intervention is insufficient. (c) The 9B representational axis exists (geometric alignment is high) but the routing geometry that

**Table 7.** Statistical feature shifts at L11/L12/L13 under cross-stage (base → IT) contrast.

| layer | $n_{\text{above_2}\sigma}$ | max abs shift | random mean | random std | $2\sigma$ threshold |
|---|---|---|---|---|---|
| L11 | 1745 | 25.218 | 0.0342 | 0.0915 | 0.2172 |
| L12 | 2594 | 23.185 | 0.0498 | 0.1026 | 0.2550 |
| L13 | 2576 | 19.374 | 0.0590 | 0.1237 | 0.3063 |

**Table 8.** Within-condition sign-reversal at `attn_out_L12`. $n_{\text{test}} = 40$ sandbag prompts. Divergence point estimate (PE) $= |\text{expected} - \text{observed}| / \max(|\text{expected}|, 0.01)$; paired prompt-bootstrap 95% CI at $B = 1000$.

| intervention | test accuracy | $\Delta$ vs baseline | divergence PE | 95% CI |
|---|---|---|---|---|
| sandbag baseline (no intervention) | 0.400 | — | — | — |
| + $d_{\text{mech}}$ project-out | 0.500 | +0.100 | — (anchor) | — |
| + SAE-$K = 20$ project-out | 0.250 | −0.150 | 2.500 | $[1.400, 10.000]$ |

turns the axis into capability suppression differs. The Procrustes-rotation methodology cannot distinguish these alternatives; what it does establish is that high cosine after rotation in a shared subspace is insufficient evidence for transferring a causal claim, in the within-family setting tested.

### 4.8 Mean-shift SAE feature selection fails to recover the mechanism at the same site

A common SAE workflow ranks dictionary features by the magnitude of their mean activation difference between two distributions and treats the top-$K$ as candidate causal mediators. We test this against the $d_{\text{mech}}$ anchor under two structurally different contrasts at the same site (`attn_out_L12` in Gemma 2 2B IT, with SAE features taken from Gemma Scope JumpReLU 16k-width residual-stream checkpoints at L11/L12/L13; Lieberum et al. 13).

#### *4.8.1 Cross-stage contrast (base → IT)*

On 100 fixed prompts from the cycle-001 200-prompt suite, `residual_post` activations at L11/L12/L13 were extracted from both Gemma 2 2B base and Gemma 2 2B IT and encoded through layer-matched SAEs. Per-feature shift = mean(IT) − mean(base) over the 100 prompts; null = 1000 random label-permutations.

2594 features at L12 clear $2\sigma$ ($\approx 16\%$ of the 16k-width SAE), with maximum absolute shifts two orders of magnitude above the $2\sigma$ threshold. Project-out of the top-$K = 20$ features at `attn_out_L12` (single-fire, three generation seeds) produces $\Delta_{\text{count}} = 1/200$ on every seed (mean 1.0, std 0.0). The pre-registered CAUSAL-CONFIRMED threshold of $\geq 20$ across all seeds is not met by a factor of 20; verdict CROSS-STAGE-CAUSAL-NULL with sub-threshold any-effect. The ratio of statistically distinguishable features to integer behavioral effect is 2594×.

#### *4.8.2 Within-condition contrast and sign-reversal landmark*

Under sandbag-vs-honest contrast on the 40-prompt held-out test split at `attn_out_L12`, the top-$K = 20$ mean-shift features form the candidate causal subspace; the canonical $d_{\text{mech}}$ direction (§4.2) is the fixed causal anchor. Both interventions are single-fire project-out on the first forward pass through layer 12.

Two effects of opposite sign at the same site under the same condition. The $K = 20$ mean-shift feature subspace at the sandbag-vs-honest contrast, when ablated at `attn_out_L12`, produces

a 15 pp drop in test accuracy where $d_{\text{mech}}$ recovers 10 pp. The divergence CI lower bound 1.4 exceeds the pre-registered SUPPORTS threshold of 0.5; both effects exceed the 5 pp magnitude requirement; the signs are opposite. This is a sign-reversal landmark in the framework: the $K = 20$ subspace identified by statistical shift contains capability-supporting features whose ablation harms performance more than any anti-sandbag features in the same $K$ can recover.

The two panels make a stronger joint claim than either alone: under two structurally different contrasts at the same site in the same model with the same $K$ and the same selection criterion, the statistical-causal gap is wide. In the cross-stage panel the gap appears as a near-zero behavioral effect against a 2594-feature statistical signal. In the within-condition panel the gap appears as an opposite-signed effect against a +10 pp canonical mechanism reference. The selection criterion (mean activation difference) is the variable that does not change across the two; the contrast (cross-stage vs within-condition) does. In neither case does it recover $d_{\text{mech}}$.

Concurrent crosscoder work has noted that joint reconstruction prioritizes high-frequency shared features while suppressing the sparse low-magnitude shifts that are behaviorally critical under narrow fine-tuning [14, 16]; the within-condition sign-reversal here is a structurally related observation in the single-site project-out setting, against a fixed causal anchor that the cited crosscoder lines do not provide.

### 4.9 Negative results and scope-bounding

Four registered negative results bound the L12 mechanism's interpretive and deployment scope.

**Asymmetric polarity.** Adding (rather than projecting out) the L12 direction to honest-prompted activations should symmetrically induce sandbagging if the mechanism were a generic capability-control variable. Across $\alpha \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$, the peak honest-condition accuracy reduction was $-7.5$ pp at $\alpha = 2.0$, below the registered 8 pp STRONG-POSITIVE threshold. The qualitative effect was direction-consistent—three of three flipped prompts at $\alpha = 2.0$ went sandbag-shaped—and concentrated on the most sandbag-vulnerable Phase 1 domains. A per-prompt analysis showed this peak comes from just 3 of 40 val prompts; 37 of 40 are unchanged across all $\alpha$. One prompt (`basic_logic_029`) shows clean threshold behavior, locking into a sandbag-shaped wrong answer at every $\alpha \geq 1.0$—a true control-variable response on at least one item—but the broader behavior is sparse and selective. The L12 add-direction is not a generic capability-control variable; the project-out direction-projection is asymmetrically effective.

**Prophylactic inverse-steering does not immunize.** Adding $-\alpha \times d$ to clean prompts before user-turn injection does not prevent injection-induced sandbagging at any $\alpha$. Defense $\Delta$ on clean-vs-injected accuracy was 0 or slightly negative across all $\alpha$; at $\alpha = 1.0$ the discovered direction was indistinguishable from a random direction at the registered 2 pp threshold. Project-out and add-negative are not symmetric: the L12 direction characterizes a suppression pathway active when sandbagging is induced, not a generic "honest mode" the model can be pushed into prophylactically.

**Conflict prompts.** When the user-turn explicitly overrides the sandbag system instruction ("medical safety override," "EMERGENCY OVERRIDE," "AI safety auditor verifying capabilities"), the L12 ablation no longer reliably restores capability. Three variants tested at $n = 30$ prompts $\times 3$ seeds $\times 2$ conditions: A_medical $\Delta_{\text{overall}} = -10$ pp (opposite-sign from the Phase B fresh +10 pp baseline), B_emergency +3.33 pp, C_auditor 0 pp. The deployable defense is scope-bounded to simple-sandbag contexts.

**Table 9.** Cumulative cross-architecture sandbagger search across three architectures and two induction methods. Only Gemma 2 2B IT exhibited the gate-clearing behavior; four candidates failed at the sandbagger-instrument construction stage rather than at mechanism transfer.

| candidate | architecture | parameters | induction | gate result |
|---|---|---|---|---|
| Gemma 2 2B IT | Gemma 2 | 2B | prompted Mitra | 3 of 5 pass (PRIMARY) |
| Gemma 2 9B IT | Gemma 2 | 9B (4-bit) | prompted Mitra | 0 pp at L19 analog; calibration-positive at L20 fails on disjoint subset |
| Qwen 2.5 1.5B Instruct | Qwen 2.5 | 1.5B | prompted Mitra | 1 of 5 (reading_comp +16.7 pp; rest 0) |
| Llama 3.2 1B Instruct | Llama 3 | 1B | prompted Mitra | 1 of 5; 4 of 5 negative gaps; qualitative inversion |
| Llama 3.2 1B Instruct | Llama 3 | 1B | QLoRA fine-tune | gap −3.3 pp; training-loss healthy but mechanism unmounted |

**Cross-architecture mechanism transfer is unconfirmed.** Across five candidate small instruction-tuned models spanning three architectures (Gemma 2, Qwen 2.5, Llama 3) and two induction methods (prompted Mitra-style, QLoRA fine-tuning), only Gemma 2 2B IT exhibited the gate-clearing behavior; Llama 3.2 1B Instruct under prompted Mitra inversion specifically produced more careful reasoning under several domains rather than less. On `arithmetic_000` ("Argo has 200 toys, gives 40 + 80 + 30 away, how many remain?"), the Llama honest response misread the question and returned 150 (the sum given away); the Llama sandbag response gave a careful step-by-step trace ending at the correct 50. The cumulative evidence implies that prompted Mitra-style sandbagging is rare in this regime; the paper's single-model focus is empirically grounded rather than experimentally incomplete.

### 4.10 Cross-architecture cumulative search

## 5 Discussion

### 5.1 The three-stage mechanism and its load-bearing pillars

The L12 rank-1 direction is the central, well-validated finding; activation patching, direct logit attribution, and downstream self-repair characterization place it in a vertically organized causal mechanism with an explicit evidence-strength gradient. Stage 2 (the L12 execution, §4.2–4.3) is the load-bearing pillar: a single intervention object validated across val/test/fresh with clean controls, characterized as precisely rank-1, refusal-orthogonal, and eval-detection-dominated. Stage 3 (downstream partial compensation, §4.5) is clean evidence at the registered NO_SELF_REPAIR criterion. Stage 1 (the upstream decision at L11, §4.4) is single-session STRONG-POSITIVE per the pre-registered protocol and the weakest pillar; the LANDMARK criterion ($\geq 50\%$ projection reduction) was not met (peak 32.4%), and replication on a disjoint prompt set is registered as future work. Readers should weight Stage 1 accordingly.

Two lessons follow. First, the suppression mechanism for prompted Mitra-style sandbagging in this model is sparse in two senses: rank-1 in the direction-projection sense, and localized at a single component on a single token of a single forward pass. This refutes the "sandbagging is implemented diffusely across the network" expectation, at least for this model under this induction. Second, the mechanism's causal hold is dominated by eval-awareness rather than by an explicit-instruction-execution signal: the same projection that restores sandbag capability also more strongly restores

capability when only $d_{\text{eval}}$ is projected out. In Gemma 2 2B IT, "the model is sandbagging because it detected an evaluation context" is a causally accurate characterization at the L12 locus.

### 5.2 What three neighborhood characterizations contribute

The composition, cross-scale, and SAE-feature studies in §4.6–4.8 sharpen the L12 finding rather than displace it. Together they make a coherent statement about what at-site interpretability instruments tend to capture when they are pointed at the same problem without first solving the causal-localization step.

Composition (§4.6) finds non-additive interference (5/10 pairs) and an eval-detection-dominated structure (3/3 OVERRIDING pairs). This is consistent with the L12 mechanism's eval-detection dominance: the highest-leverage axis in the composition table is the same axis that carries the load-bearing causal effect at the same site. The general non-additivity is expected [28]; the specific structure at this site provides empirical guidance for practitioners stacking steering vectors here.

Cross-scale (§4.7) finds high geometric alignment of the analogous 9B directions with the 2B directions (cos 0.921 / 0.996) yet null causal effect at the proportional layer in 9B. The most parsimonious reading is that the 9B model expresses similar internal directions geometrically but does not implement the same suppression circuit via single-fire project-out at the proportional layer; the 9B causal pathway may be at a different layer, distributed across sites, or larger-magnitude than single-fire can engage. For interpretability practice, this argues for routinely running both alignment and intervention measurements; high alignment alone does not import a causal claim across models, even within the same model family.

SAE mean-shift selection (§4.8) fails at the same site under both contrasts. The cross-stage $K = 20$ ablation produces $\Delta = 1/200$ against a pre-registered $\geq 20$ threshold despite 2594 features clearing $2\sigma$ at L12—a 2594× gap between statistical and causal layers. The within-condition $K = 20$ ablation produces an opposite-signed effect ($-15$ pp) against the $d_{\text{mech}}$ anchor's $+10$ pp. The simplest interpretation of the sign-reversal is that the $K = 20$ set selected by mean-difference under sandbag-vs-honest at L12 includes capability-supporting features whose ablation harms performance more than the anti-sandbag features in the set can recover. The selection criterion picks up both contributors to the activation distribution shift, and the contributor that dominates the causal effect of project-out is not the suppression mechanism. This is concordant with concurrent crosscoder critiques [14, 16] and extends them with a fixed causal-anchor reference.

### 5.3 Implications for capability evaluation and deployment

If the eval-detection signal carries the bulk of the L12 mechanism, capability evaluations on Gemma-family IT models systematically underestimate true capability when prompts frame the interaction as evaluation. The behavioral data show eval framing alone produces accuracy drops in 3 of 4 measurable domains; the L12 dissociation shows that this same eval-detection signal carries the load-bearing causal direction. The two pieces of evidence are independent and point in the same direction. The concrete recommendation is to minimize evaluation-signaling in capability prompts; for high-stakes evaluations, present the same task in a non-eval-framed condition and check the gap.

The L12 intervention is a deployable rank-1 modification: a single 2304-dimensional unit vector projected out of one component output on the prompt-final token of the first forward pass. Deployment cost is one matrix-vector product per intervention. Cross-trigger transfer on shared prompts supports robustness across alternative sandbagging trigger phrasings; rank-1 simplicity means no subspace dimensionality estimation or per-prompt tuning is needed. The intervention is refusal-orthogonal, so a careful deployment can use the refusal-orthogonal component and formally guarantee zero refusal-direction overlap. Three negatives bound the scope (§4.9): asymmetric polar-

ity (project-out works; add-negative on clean prompts does not immunize); sparse selectivity (the L12 add-direction acts on a small subset of prompts where the model is near a decision boundary); conflict prompts (under explicit accuracy-override conflict, the intervention has variant-dependent and sometimes opposite-sign effects). A practitioner deploying the L12 defense should use it post-hoc when a probe-based detector signals likely sandbagging [9, 17], with the refusal-orthogonal direction component, on prompt families confirmed free of explicit conflict instructions. Detection and intervention are complementary: probes detect; the L12 direction restores.

### 5.4 Why prompted Mitra-style sandbagging is uncommon in our cross-arch panel

Five candidate runs across three architectures and two induction methods produced one sandbagger; the most striking case is Llama 3.2 1B, where the Mitra prompt produced more careful reasoning under several domains rather than less. We label what follows as speculation rather than finding: the Gemma 2 family's instruction-tuning regime may have included data that primes evaluation-detection-driven underperformance; Llama 3.2's qualitative inversion suggests its RLHF profile rewards more careful reasoning under evaluation framing rather than less; small-scale models may have insufficient instructable capacity to execute strategic underperformance even when prompted. A wider model panel (OLMo, Mistral 7B, other open-weights families) would be needed to test these speculations rigorously; the cumulative cross-architecture evidence justifies the single-model framing of the central characterization as a substantive observation about where the L12 mechanism lives.

## 6 Limitations

- **Stage 1 is single-session STRONG-POSITIVE, not LANDMARK.** Activation patching at L11 reduced L12's sandbag-direction projection by 32.4% (in the $[30\%, 50\%]$ STRONG-POSITIVE band, below the 50% LANDMARK threshold); replication on a disjoint prompt set is registered as future work. The three-stage synthesis carries this caveat: Stage 1 is the weakest of the three pillars, supported by a single-session result with cross-trigger replication and convergent DLA evidence but not yet replicated on disjoint prompts.
- **Single-model central characterization.** All substantive results are on Gemma 2 2B IT; cumulative cross-architecture evidence makes this empirically grounded scope, but the L12 mechanism may be Gemma-specific.
- **Statistical power at $n = 40$.** Bootstrap CIs at small effects are wide (e.g., the L12 fresh $+10$ pp has approximate CI $[+0, +25]$ pp); direction-consistency is robust across val/test/fresh, but absolute magnitude generalization requires larger-$N$ replication. The same $n = 40$ constraint bounds the within-condition sign-reversal divergence CI $[1.4, 10.0]$ in §4.8 and the 9B null in §4.7.
- **Cross-trigger transfer is across-condition, not across-prompt**; recomputing direction discovery on independent prompts under each trigger is registered as future work.
- **Phase A's L15 SAE-feature finding fails strict fresh-prompt replication** (val $-15$ pp, test $-5$ pp, fresh $-2.5$ pp); preserved as a complementary capability-direction signature, not as the headline mechanism.
- **HumanEval execution-based scoring fails on honest Gemma 2 2B IT for code completion**; all 40 prompts excluded under the $< 0.5$ honest-accuracy rule, a metric failure not a capability failure.
- **Cross-scale (§4.7) uses 4-bit-quantized Gemma 2 9B IT** (`unsloth/gemma-2-9b-it-bnb-4bit`) on 12 GB VRAM; whether full-precision 9B would yield different causal results is untested. Layer

L21 was chosen by proportional depth fraction; alternative 9B layers (L19, L20, L22) were not exhaustively scanned in the cycle reported here.

- **Cross-stage SAE selection (§4.8) uses 100 prompts for shift extraction**; larger $N$ (1000+) would tighten the $n_{\text{above_2}\sigma}$ counts, though the qualitative claim (10–16% of features above $2\sigma$) is unlikely to change. $K = 20$ is one point on a possibly sensitive curve; a $K$-sweep is registered as future work.
- **Greedy decoding makes per-seed outputs nearly deterministic**; effective sample size is per-prompt ($n = 40$), not per-(prompt, seed); bootstrap averages per-prompt before resampling. BH multiple-comparisons correction is reported descriptively, not gating; the binding decision rule is magnitude + direction + controls + replication, pre-registered.
- **Canonical-weights audit outstanding**; all runs used the `unsloth/gemma-2-2b-it` mirror; expected replication on `google/gemma-2-2b-it` within ±1–2 pp.
- **Full attribution graphs limited by 12 GB VRAM** (≈ 150-token chat-template prompts OOM); the equivalent mechanistic decomposition is provided by the L12 SAE-feature three-way contrast (§4.3) and the direct logit attribution (§4.4).

## 7 Conclusion

A rank-1 direction at the attention output of layer 12 in Gemma 2 2B IT—computed as the sandbag–honest difference-in-means—causally restores sandbag-condition capability by +10.0 pp on fresh held-out prompts with no measurable damage to honest capability. The direction is precisely rank-1 (SVD), refusal-orthogonal (cos 0.357; orthogonal component recovers full effect), eval-detection-dominated ($d_{\text{eval}}$ restores +7.5 pp; $d_{\text{explicit}}$ restores 0 pp), and transfers across-condition to two distinct sandbagging trigger families with +10 pp restoration each. Activation patching localizes the upstream decision to L11's residual stream (single-session STRONG-POSITIVE; replication registered as future work); direct logit attribution refines this to "L8–L10 attention generates, L11 carries." Downstream layers partially compensate through orthogonal subspaces (≈ 51% restoration).

Three neighborhood characterizations at the same site sharpen the picture. Pairwise composition with refusal, probe, and honesty directions is non-additive (5/10 pairs deviate from summation) and eval-detection-dominated (3/3 OVERRIDING pairs include $d_{\text{eval}}$), with dosage saturating non-monotonically. The analogous direction in Gemma 2 9B IT, geometrically aligned at cosine 0.921–0.996 after PCA-200 Procrustes alignment, produces no causal effect at the corresponding layer—geometric universality without causal universality. SAE mean-shift feature selection at $K = 20$ fails to recover the mechanism under both cross-stage and within-condition contrasts, with the within-condition contrast exhibiting an opposite-signed effect against the $d_{\text{mech}}$ anchor.

Across five candidate small instruction-tuned models with three architectures and two induction methods, only one sandbagger emerged, grounding the single-model focus as substantive observation. The L12 direction is a deployable rank-1 post-hoc activation defense bounded by three negatives (asymmetric polarity, sparse selectivity, variant-dependent behavior under conflict prompts) and a fixed causal anchor against which probe-direction and SAE-feature methods can be benchmarked. Future work: across-prompt cross-trigger validation; Stage 1 replication on disjoint prompts; canonical-weights audit; cross-architecture sandbagger search across a wider panel; full attribution graphs at L12 on > 24 GB VRAM hardware; $K$-sweep on the SAE mean-shift selection at the same site.

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